

Collaboration, Organizational Capacity, and Sustainability: The Cross-Sector Governance of Smart Cities

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Abstract

Interactions with other organizations likely influence how local governments confront complex problems. Drawing on theories of collaborative governance, we investigate the extent to which collaboration affects U.S. local governments' prioritization of the sustainability benefits of Smart Cities Technologies (SCT). Using data from the International City/County Management Association (ICMA) 2016 Smart Cities Survey and the U.S. Census Bureau, we find that collaboration with local, regional, and private sector partners influences the extent to which local governments prioritize the sustainability benefits of SCT. Organizational capacity is also an important predictor of perceived benefits of SCT. Our findings suggest a hidden side of boundary spanning, suggesting that collaboration can enhance opportunities for sustainability technologies.

Keywords

sustainability, collaborative governance, local government, capacity, smart cities technologies (SCT)

Practitioner Points

1. As local governments face increasingly complex administrative problems, collaboration is often seen as a way to secure needed resources, gather information, and build expertise. However, collaboration can also influence a local government's priorities, as it offers exposure to new perspectives and different solutions.
2. Smart Cities Technologies may provide local governments support as they work to address complex problems. Increasingly, cities are investing in Smart Cities Technologies to create systems to help administer sustainable, innovative, and efficient local programs.
3. Cities that report collaborating often are more likely to see benefits of Smart Cities Technologies, indicating the process of interacting with other localities influences perceptions of the value and benefits of new technology.
4. Capacity also influences how local governments perceive Smart Cities Technologies. Cities interested in pursuing Smart Cities Technologies should garner support from elected and appointed leaders and work to improve their government's financial stability. Procurement and project management capacity are not currently important capacity barriers for local governments.

Introduction

Contemporary public organizations regularly confront wicked problems (Sørensen & Torfing, 2021) such as climate change, natural disasters, sustainability, and pandemics such as the COVID-19 pandemic. Wicked problems are

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multilevel, multi-actor, and multi-sectoral challenges (Bianchi et al., 2021), and addressing them requires accessing and leveraging the expertise and capacity of various interdependent stakeholders to form collaborative and strategic partnerships (Bianchi et al., 2021; Klijn & Koppenjan, 2000). These strategic relationships form collaborative governance wherein organizations develop boundary-spanning processes and procedures to constructively address public problems (Emerson et al., 2012). Collaborative governance has been widely applied to some of the most complex public problems of our time and has a demonstrated ability to improve community outcomes (Brink & Wamsler, 2018; Lubell & Lipert, 2011; McGuire & Silvia, 2010; Scott & Thomas, 2017).

Climate change presents a wicked problem for local governments. In part this is due to inaction from national and international governments (Schwartz, 2019; Selin & VanDeever, 2011) but is also due to unique local capabilities (Acuto, 2013; Hölscher et al., 2019). Because of this many local governments have taken the lead in planning for and implementing sustainable policies and programs. As environmental, economic, and social systems experience foundational threats, local governments are increasingly taking steps to preserve resources and institutions for future generations (Fiorino, 2010). In addition to strengthening political and social systems (Zeemering, 2018), local governments have been at the forefront in reducing waste (Warbroek et al., 2019), incorporating green technology policies (Terman & Smith, 2018), incentivizing sustainable economic investments and growth (Laurian et al., 2017), and improving communities' environmental and social conditions (Ji & Darnall, 2018).

Many municipalities have committed to becoming so-called "smart cities," which are cities that use "information and communication technology to enhance livability, workability, and sustainability" (Smart Cities Council, 2015, p. 6). Thus, a smart city leverages technology to create a community that is sustainable, competitive, and with increased quality of life (Bakıcı et al., 2013). A large part of every smart city initiative is active collaboration between governments, private firms, residents, and others who have critical resources and knowledge to contribute to a city's structure and ability to deliver public services (Kim, 2018; Lam & Yang, 2020; Pianezzi et al., 2021; Viale Pereira et al., 2017). Although much has been written about smart city initiatives, few studies have investigated how the act of collaboration affects smart cities. Collaboration is known to influence organizational strategy, as the collaborative process can introduce new ideas, alter perceptions about priorities, and restructure the allocation of resources (Hartley et al., 2013; Vangen et al., 2015). Building on these studies, we investigate how collaboration influences the prioritization of sustainability benefits associated with Smart Cities Technologies (SCT).

SCT are a worldwide phenomenon. As a result, we see many SCTs being adopted and implemented in Europe and Korea (Komninos et al., 2013; Lim & Rasul et al., 2022; Trindade et al., 2017; Yigitcanlar, 2015). Until now, studies about the diffusion and adoption of SCT have largely focused on contextual variables, such as size, form of government, political leanings, and managerial constraints (Angelidou, 2017; Araral, 2020; Snis et al., 2021). Yet, smart cities approaches are also likely the result of managerial factors, including organizational support, capacity, and collaborative influences (Araral, 2020; Viale Pereira et al., 2017). Accordingly, in the context of the United States, we use collaborative governance theories to make sense of the managerial factors that contribute to achieving sustainability benefits, with a particular focus on the role of collaborative partnerships and local-level governance capacity.

Based on an analysis of data from the International City/County Managers Association (ICMA), our findings suggest that collaboration—local, regional, and across sectors—influences the commitment to becoming a smart, sustainable city. We also find that capacity is critical, especially leadership and familiarity with forming cross-sectoral partnerships. Cities that engage in boundary spanning activities and build mutually beneficial partnerships appear to be better situated to develop and implement technology-driven programs to promote sustainability and address wicked problems.

This research contributes to the understanding of how collaboration affects local governments. Confronted with complex problems and uncertainty, collaboration offers localities the chance to gather information, learn from each other, and identify innovative policy solutions. Collaboration may be a way to reduce the transaction costs associated with learning about new technologies (Dyer, 1997), offer exposure to innovative policy solutions and insights into their implementation (Jukić et al., 2019), and may even shift the values of local governments (Eriksson et al., 2020). However, collaboration is not the only factor that matters. Organizational capacity also influences a government's willingness and ability to adopt new policy solutions (R. K. Christensen & Gazley, 2008). Organizations with substantial capacity barriers, such as unsupportive leaders or insufficient finances, are less likely to perceive sustainability benefits of SCT, suggesting that collaboration cannot necessarily overcome the realities that public managers face inside their own organizations. Our research indicates that well-resourced localities with supportive leaders who collaborate within their region and across sectors are likely to perceive SCT as a path toward improving sustainability. Organizations without these characteristics are likely not to see (or value) sustainability benefits as much.

The remainder of the article is organized as follows. In the next section, we define sustainability and introduce smart cities initiatives, setting the stage for local government actions and how these efforts are designed to improve sustainability. We then review the literature on collaborative governance and organizational capacity, using these lines of scholarship to theorize about the factors that influence smart cities initiatives. Following the literature review, we introduce the data and analytical methods used in this study and then present our findings. The article concludes with a discussion of the implications of this research for public management theory and practice.

Smart Cities Initiatives and Sustainability

While researchers and practitioners use various definitions for sustainability, this article adopts the United Nations' definition of sustainable development which is: "*Development that meets the needs of current generations without compromising the ability of future generations to meet their needs and aspirations*" (WCED, 1987, p. 43). The scholarship on sustainability has evolved over the years to encompass aspects of social and governance systems, policy and program goals, and administrative components (Bell & Morse, 2003; Fiorino, 2010; Krause et al., 2016; Leuenberger & Bartle, 2014). Scholars have identified three primary pillars of sustainability: social sustainability, economic sustainability, and environmental sustainability (Estoque & Murayama, 2014; Moldavanova, 2016; Opp & Saunders, 2013; Pettibone, 2016; X. Wang et al., 2012). Similarly, ICMA views local sustainability as a holistic approach that looks at how local government leadership can capitalize on the connections between economic, social, and environmental sustainability to create thriving and livable communities (ICMA, 2022).

By adopting and implementing ICT infrastructures to support social and urban development through improving the economy, citizen participation, efficiency, environmental protection, living conditions, etc. (Bakıcı et al., 2013; Przybilowicz, 2022), smart cities can encompass a range of strategies that address all three pillars of sustainability. Sustainability goals are a new set of responsibilities facing governments in an environment of population growth, resource depletion, and environmental degradation. In the past, local governments may not have had the incentive or the capacity to achieve sustainability goals (Krause et al., 2023), but that is no longer the case. Due to changes in demographics, local governments are now important stakeholders in sustainability advancement and are increasingly interested in economic development, cost savings, and improving quality of life (Krause et al., 2023).

One of the tools to achieve sustainability goals is the adoption of SCT (e.g., Lim et al., 2019). A city can be considered to be a system of systems consisting of various interconnected objects (Schipper & Silvius, 2018). Smart cities focus on environmental sustainability, improving quality of life, enhancing knowledge, building intellectual capital, and citizen participation in government (Dameri & Cochhia, 2013).

The "smart cities" innovation dates to the Smart Growth Movement of the late 1990s (Höjer & Wangel, 2015), but the idea predated that time and may have even been considered in the 1960s (Gabrys, 2014). However, it gained particular attention beginning in the 2010s, building momentum as various urban centers raced to integrate new technologies to enhance planning, growth, interconnectedness, and sustainability (Schipper & Silvius, 2018). In 2015, the U.S. Department of Transportation's Smart City Challenge encouraged local governments to adopt technology solutions to improve efficiency and systems integration (Lehr, 2018). Smart technologies are also increasingly being used in other policy areas, including parks, law enforcement, and education (Kumar et al., 2020).

Generally, SCT can positively impact the social, economic, and environmental aspects of a city (Lim et al., 2019). Benefits are manifold as "smart cities function as agents of strategic transformation and provide resources and pathways for increasing operational efficiency, competitiveness, quality of life and for approaching sustainability" (Sharifi, 2020, p. 1270). Through a combination of human efforts and technology, SCT contribute to sustainability goals such as addressing urban problems of the environment (e.g., reducing carbon footprint), the economy (e.g., efficient use of resources), social programs (e.g., housing, transport, etc.) (Hollands, 2015), and social inclusion through increased access to government services to those who may have had difficulty in accessing them earlier (Neirotti et al., 2014).

Researchers have also found a positive impact of smart cities on economic growth (Caragliu & Del Bo, 2019), economic development (Abutabenjeh et al., 2022), and government revenue performance (Awasthi et al., 2021; Nukpezah et al., 2022). There is also qualitative evidence that smart cities lead to sustainable development (Alfaro Navarro et al., 2017; Angelidou, 2015; Dameri & Ricciardi, 2015; Hollands, 2015; Lim et al., 2019; Neirotti et al., 2014; Sajhau, 2017; Trindade et al., 2017; Wiig, 2015; Yigitcanlar et al., 2018).¹ For many local governments, SCT are central elements of their sustainable development programs for the 21st century.

There are valid concerns raised in the literature about the potential downsides of SCT. Some concerns related to SCT are privacy and security issues, polarization and inequality, and diminishing freedom of speech and democracy (Lim et al., 2019). These concerns, however, are not inherent in SCT themselves but rather a failure in how they are designed and implemented (Okafor et al., 2022; Clever et al., 2018). When SCT are developed with a focus on mitigating these concerns, they can help cities become more efficient, sustainable, and livable. Therefore, concerns about SCT should not discount the benefits of SCT. Though there may be an adverse impact of SCT (Burns & Andrucki, 2021), additional empirical research is needed on the role of contextual factors and their potential influence on the benefits of SCT.

Literature Review: Collaborative Governance, Organizational Capacity, and Sustainability

Addressing wicked problems like sustainable development requires accessing and leveraging the expertise of various interdependent stakeholders to form strategic partnerships and collaboration (Bianchi et al., 2021; Klijn & Koppenjan, 2000), as well as the organizational capacity to adopt and effectively manage complex initiatives and partnerships (Swann & Deslatte, 2019; X. Wang et al., 2012). Smart cities initiatives, which are one potential solution for sustainable development, arise from interorganizational collaboration, information sharing, and complex incentive programs (Lehr, 2018). Though there is ample evidence of the benefits of sustainability and smart cities approaches, to this point, few studies have investigated the extent to which collaborative governance and organizational capacity influence local government sustainability programs. To fill this gap, this research analyzes the impact of collaboration and critical capacity to determine what motivates (or prevents) local governments as they pursue sustainability goals.

Existing research shows that collaborative governance can be an effective way to address wicked problems (Daviter, 2017; Head & Alford, 2015; Roberts, 2000). However, little is known about how collaboration may affect the prioritization of sustainability as a policy objective. As Head and Alford (2015) note, collaboration can lead to a better definition of wicked problems, aid in the generation of better or more practical policy solutions, and improve efforts to implement new programs. Accordingly, it is likely that collaborative governance may have benefits for sustainability initiatives.

We adopt Emerson et al. (2012) definition of collaborative governance which holds that collaboration is “*the processes and structures of public policy decision making and management that engage people constructively across the boundaries of public agencies, levels of government, and/or the public, private and civic spheres in order to carry out a public purpose that could not otherwise be accomplished*” (p. 2). This definition highlights three elements that are critical for our study of sustainability: boundary spanning, constructive engagement, and mutual dependence.

Because social, political, economic, and environmental problems span the boundaries of traditional organizations, individual organizations, including cities, cannot work toward sustainability goals alone. Single organizations can rarely address even small pieces of these problems independently. Governments must work together, partnering with both horizontal and vertical counterparts to exchange sustainability information and other resources (Abels, 2014). They must also work closely with private sector firms through a combination of regulation and partnership to incentivize behavior change and access the latest technologies and expertise (Klein, 2020). As a result, boundary spanning is likely to be an essential element of the adoption and implementation of sustainability programs.

Similarly, sustainability efforts are often dependent on the actions of partners. For example, local governments committed to reducing their environmental impact will struggle to make meaningful progress unless other jurisdictions make similar commitments (Soujaa et al., 2021). The success of local initiatives may also depend on the provision of resources from public and private sector partners, who have access to funding, technology, and capacity that are necessary for local-level success (Bromberg & Manoharan, 2015; Fiorino & Weted, 2020; Whitford & Wong, 2009). Resources may come from national, regional, or even internal (cross-departmental) partners, as well as private and nonprofit organizations (Benavides & Nukpezah, 2020; Y. Kim et al., 2020). Depending on the criticality of the need and the extent to which organizations can offer up their own resources in exchange, mutual dependence will arise, wherein the pursuit of resources forms the basis for the allocation of power (Malatesta & Smith, 2014; Pfeffer & Salancik, 1978). This type of dependence, though laden with risks, is a distinguishing feature of collaborative governance and can secure needed resources while building commitment between partners to the exchange.

Finally, collaborative governance involves partners working together to address a common challenge, which offers them the chance to create something that they would not be able to make or find on their own (Abutabenjeh et al., 2023; Diallo & Reid, 2021; Emerson et al., 2012; Johnston & Kouzmin, 2010). For sustainability initiatives, this

includes combining policy and management commitments with innovative technologies (Park et al., 2021; Yi et al., 2018). Ultimately, sustainability efforts seek to change the behaviors of residents, administrators, political actors, and firms to protect valuable resources and systems for future generations. This kind of effort requires constructive engagement where partners from a range of backgrounds bring their talents, resources, and beliefs together to generate something new and lasting for a community of people.

Collaborating is risky (Thomson & Perry, 2006), but it has great benefits if done well (Ansell & Gash, 2007). For cities interested in sustainability, different types of collaboration will be necessary to span boundaries, develop mutually beneficial partnerships, and build new resources. Internally, cities need to work to establish strategic objectives, coordinate efforts, and implement programs. Locally, cities must work to gain commitments from other jurisdictions, access innovative technologies, and understand the priorities of residents (Abels, 2014). In addition, they must work with state and local partners to identify and secure resources and a supportive policy environment. We hold that cities that are actively engaged in collaborative initiatives like these will be more likely to prioritize sustainability, especially as a part of their smart city initiatives. Accordingly, we hypothesize:

H1a: Local governments that collaborate with local partners are likely to perceive higher sustainability benefits of SCT.

H1b: Local governments that collaborate with state and national partners are likely to perceive higher sustainability benefits of SCT.

H1c: Local governments that engage in public-private partnerships are likely to perceive higher sustainability benefits of SCT.

Organizational Capacity and Sustainability

Working toward sustainability goals requires organizations to unlearn previous behaviors and develop different types of capacity to implement new responsibilities (X. Wang et al., 2012). Organizational capacity refers to the ability of an organization to carry out its mission and reach its goals (Koerner & Johnston, 2021; X. Wang et al., 2012), given the extent to which an organization possesses the necessary resources and skills. Gargan (1981) notes that “program failures are frequently attributed to an agency’s lack of sufficient capacity” (p. 649), suggesting that a lack of capacity can pose a problem for organizations seeking to engage in complex intergovernmental partnerships. Similarly, Van Slyke (2003) finds that “the cost of developing capacity is a barrier for many public agencies, especially at the county level,” (p. 301), indicating the importance of organizational capacity for cross-sectoral partnerships and contract management. Accordingly, organizational capacity and barriers to capacity may help explain how local governments prioritize the sustainability benefits of SCT. Technical, managerial, and financial capacities are essential for the attainment of sustainability goals (Liao et al., 2019; X. Wang et al., 2012) and are directly related to the planning and implementation of sustainability initiatives (Swann & Deslatte, 2019).

Organizational capacity includes the skills and knowledge of employees, access to financial resources, and political (or leadership) support (Joaquin & Greitens, 2021; Swann & Deslatte, 2019). Previous studies of capacity have largely focused on financial resources as inputs to policy systems and program management (e.g., Hou et al., 2003). Yet, it is likely that both administrative and leadership capacity are important for the success of sustainability initiatives, as technical knowledge and managerial skills are essential too (Swann & Deslatte, 2019). To purchase new technologies and successfully implement them, financial investments are doubtless necessary. However, as many scholars have shown, having sufficient cash on hand is necessary but not a sufficient condition for organizational change (Fernandez & Rainey, 2017). Organizations must also have dedicated human capital and relevant managerial skills (Joaquin & Greitens, 2021; X. Wang et al., 2012). In the context of smart cities, this includes the ability to manage complex public procurement processes and project management skills to oversee the implementation of new sustainability initiatives. Accordingly, in the following sections we explore five main areas of organizational capacity (internal, procurement, project management, leadership, and financial) that are likely to affect perceptions of sustainability benefits, offering insights into the ways different types of capacity influence local governments.

Internal Capacity

Internal capacity in government refers to the “ability to marshal, develop, direct and control financial, human, physical and information resources” (Ingraham et al., 2003, p. 15). This clearly extends beyond the basic attributes of resource

acquisition and funding (Kushner & Poole, 1996) to include a range of both tangible and intangible assets. As R. K. Christensen and Gazely (2008) note, “much of the difficulty in defining organizational capacity rests in its multiple qualities, as both an input and a throughput, a resource and a process” (p. 266). They argue that organizational capacity is often viewed as an overall attribute, but it can also be analyzed as an expansive, multi-dimensional, empirical construct (R. K. Christensen & Gazely, 2008). We follow their guidance and begin with internal capacity as an overall characteristic of organizations, and then delve more deeply into specific aspects of capacity that matter in the context of smart cities.

Possessing an overarching level of capacity is necessary for the attainment of all organizational objectives (Ting, 2011; Vinzant & Vinzant, 1996). More recent studies show that internal capacity is also important for achieving sustainability goals (Deslatte et al., 2021; Hawkins, Krause & Deslatte, 2021). In their analysis of organizational effectiveness and sustainability, X. Wang et al. (2012) find that greater organizational capacity, measured as an overall perceptual assessment of internal capacity, is correlated with increases in the success of sustainability programs. Like other sustainability efforts that blend elements of technological and managerial changes with new incentives and programs, smart cities initiatives are likely to be more successful if a government has sufficient overarching internal capacity. Conversely, insufficient management capacity may serve as a barrier to partnerships and implementation (Hackler & Saxton, 2007; Van Slyke, 2003). Accordingly, we hypothesize:

H2a: Local governments with more internal capacity are likely to perceive higher sustainability benefits of SCT.

Complex Procurement Capacity

Smart cities programs require the procurement of complex goods and services (Lam & Yang, 2020; Neirotti et al., 2014). Governments are increasingly reliant on the private sector for the provision of complex goods and services (Brown et al., 2013), but successful management requires sufficient capacity for designing, overseeing, and managing the procurement process (Brown & Potoski, 2003; Brunjes, 2020; Joaquin & Greitens, 2012; Stritch et al., 2020). Smart cities are extreme examples of this reliance. To gather data to inform the daily operation of the government, smart cities purchase sensors and other devices from private firms to provide insight into various types of activity around a city (Neirotti et al., 2014). The data that these devices generate must then be analyzed to identify ways to alter or improve program implementation. In addition, updates to basic infrastructure (such as roads, electrical grids, and telecommunications) are necessary to facilitate the exchange of data and attain sustainability goals. These contractors often act more as partners, as the services they provide are critical for the locality (Pihlajamaa & Merisalo, 2021). Accordingly, smart cities are tasked with managing a large network of relational contracts (Bertelli & Smith, 2010) with firms that are providing integral services for a wide range of new and existing public programs.

To succeed in such a context, sufficient procurement capacity is necessary. As the complexity of the goods or services purchased rises, the transaction costs associated with the procurement also rise (Brown et al., 2006; Brunjes, 2020). Specific transaction costs include searching, bargaining, transferring knowledge, and overseeing providers during the period of a contract’s performance. Procurement officials are responsible for managing each phase of the procurement process and can view added complexity and ambiguity as increased costs that require procurement expertise (Stritch et al., 2020). Since much of the work on smart cities initiatives has malleable objectives, indeterminate outcomes, and hard-to-predict timelines (Lombardi et al., 2012), it is likely that procurement officials encounter highly complex procurements that require a great deal of technical and process expertise. Reforms, including efforts to improve sustainability, require implementation capacity (T. Christensen & Lægheid, 2011). Thus, organizations with lower capacity will struggle to develop and implement such initiatives (Koerner & Johnston, 2021). Consequently, the complexity of procurement may act as a barrier to engagement in these types of initiatives. Accordingly, we hypothesize:

H2b: Local governments with greater procurement capacity are likely to perceive higher sustainability benefits of SCT.

Project Management Capacity

In their modern classic on the changing nature of public administration in the era of contracting out, Milward and Provan (2000) argue that the basic nature of government work is changing. Instead of a world where states are tasked

with directly delivering public services, we now live in the era of “governance” where states are responsible for organizing resources, assembling networks, managing partnerships, and evaluating the success of projects. This requires a new set of skills from government officials. Rather than having the expertise and technical skills to deliver services directly, public administrators are instead required to be experts in managing projects that span large networks of partners, including contractors (Forrer & Kee, 2004; Milward & Provan, 2000; Voelz, 2010). Essentially, the modern-day public administrator needs to be adept at project management, including initiating new networks, establishing roadmaps, overseeing the execution of work by third parties, monitoring performance, and ensuring the successful delivery of work performed by someone other than a government worker.

Project management is an organized approach to managing complex initiatives that often span multiple organizations (Gareis, 2010; Schipper & Silvius, 2018; Turner, 1996). Project management involves planning, organizing, directing, and controlling an entity’s resources to achieve specific goals and objectives (Kerzner, 2017; Schipper & Silvius, 2018). Highly trained, carefully allocated project management capacity is essential for managing any project (Burke, 2013), but it is especially important for sustainability programs (Martens & Carvalho, 2017; Larsson, 2020). Insufficient project management expertise, personnel, and experience can be a barrier for local governments (Bjorvatn & Wald, 2018). Due to the need to establish goals and objectives, reliance on third parties (largely through contracts), and constant monitoring of performance and integration of partners, project managers and their associated skills are essential for sustainability programs like smart cities. Accordingly, we hypothesize:

H2c: Local governments with greater project management capacity are likely to perceive higher sustainability benefits of SCT.

Leadership Capacity

Organizational change is also often reliant on support from leaders (Fernandez & Rainey, 2017; Gilley et al., 2009). In the public sector, leaders often wield coercive power (Fernandez, 2004), serve as critical connections to resources (Williams, 2002), and can be effective entrepreneurs for innovative policy change (Hartley et al., 2013). For major changes such as adopting SCT for sustainable development, leaders can serve as champions that advocate for investment and political support from overhead and external sources (Agranoff, 2017), while also structuring and motivating support from critical internal assets (Bankins et al., 2017; Nowell et al., 2016).

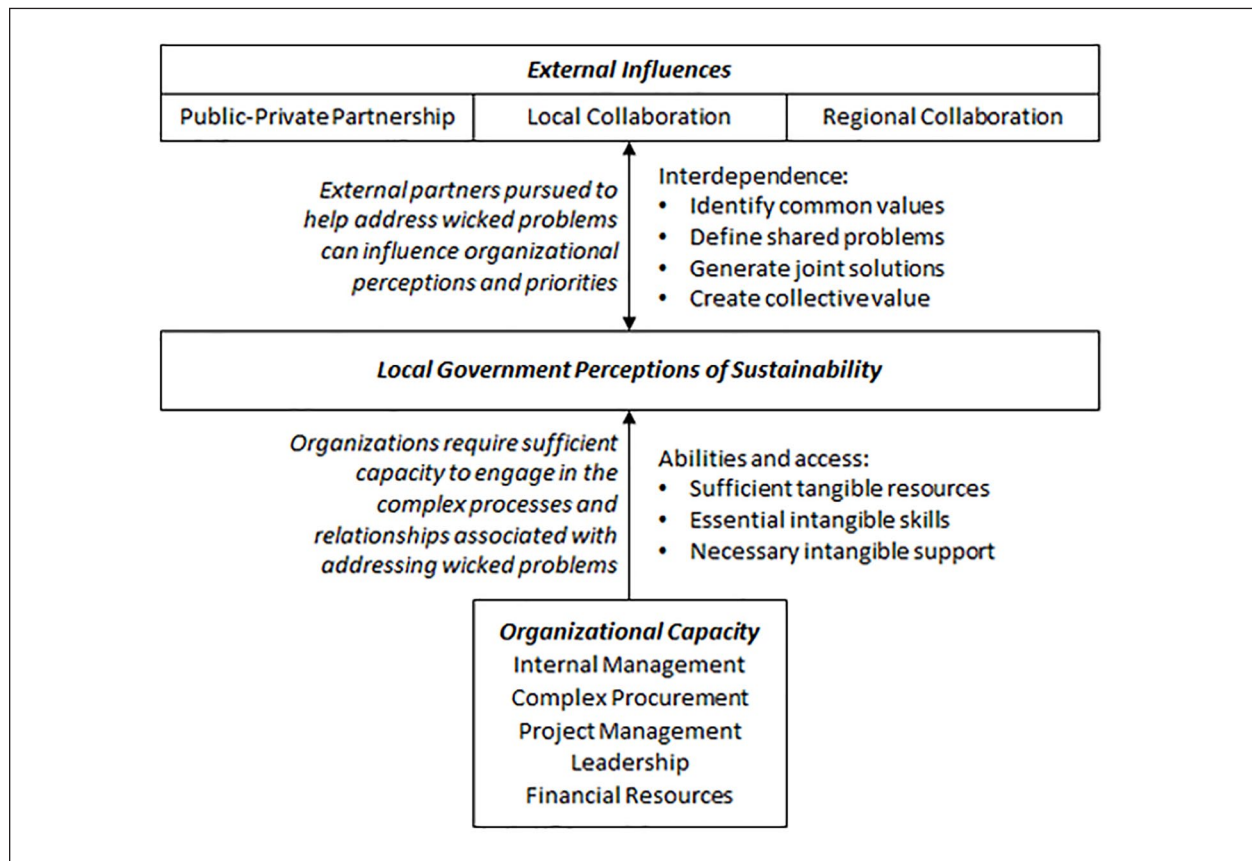
In the public sector, leadership can come from multiple sources, including elected executives and appointed managers (Svara et al., 2013), community leaders (Homsy et al., 2019), and regional and state policy leadership (Homsy & Warner, 2015). These leaders help promote sustainability in multiple ways. For example, community leaders collaborate with state and private partners (Cutter et al., 2008; J. Kim, 2018) to press for sustainability programs, while state officials can provide leadership on policy. In the context of smart cities, local government leaders can provide dedicated staff to design, promote, and manage sustainability initiatives (Homsy et al., 2019).

Studies show that strong leadership is an essential prerequisite to achieving positive results on sustainability (e.g., Nelson & Svara, 2015). Leaders have the authority to allocate resources to sustainability initiatives (Deslatte & Stokan, 2020) and can encourage local action toward sustainability (Liao et al., 2019). Conversely, a lack of leadership can be a barrier to sustainability initiatives (Saha & Paterson, 2008). An absence of effective leadership may result in the reallocation of resources away from sustainability initiatives, thereby diminishing the impact of sustainability programs (Svara et al., 2015). Thus, leadership serves as a critical resource for sustainability initiatives, and its absence is notable in its detrimental effects. Accordingly, we hypothesize:

H2d: Local governments with supportive leaders are likely to perceive higher sustainability benefits of SCT.

Financial Capacity

In addition to workforce capacities like procurement, project management, and leadership, financial resources are also necessary elements of organizational capacity. Financial capacity has received the most attention from scholars, likely due to its ease of measurement (Hou et al., 2003). To successfully implement any organizational change, having access to sufficient resources is a necessary, but not always sufficient condition (Fernandez & Rainey, 2017). Fiscal constraints have been frequently associated with lower levels of innovation (Walker & Chaiken, 1982; Schick, 1988), less

Figure 1. Theoretical framework: collaboration, organizational capacity, and sustainability

capability to respond to emerging problems (Overmans & Noordegraaf, 2014; Warner et al., 2021), and cuts to new or polarizing programs (Raudla, 2021; Q. Wang & Peng, 2018).

Sustainability initiatives like smart cities are highly innovative and polarizing (Appio et al., 2019; Batty, 2020). Sustainability programs impose substantial financial burdens during the formation (Opp et al., 2014; Saha & Paterson, 2008) and implementation stages (Hawkins et al., 2016). Due to general conditions of resource scarcity, sustainability goals compete with other organizational needs whose returns may be quantified and perceived more favorably in the short term (Hawkins et al., 2016). That is, sustainability initiatives like smart cities are likely to incur high costs, especially in comparison to other local programs, and require substantial and continued investments for success. Consequently, areas with higher financial capacity, particularly in terms of access to tangible funding, are more likely to engage in smart cities programs (Krause, 2011). Accordingly, we hypothesize:

H2e: Local governments with more financial resources are likely to perceive higher sustainability benefits of SCT.

In this section, we have demonstrated how different aspects of collaborative governance and capacity could influence the extent to which local governments perceive sustainability benefits associated with SCT. Based on this, we offer a theoretical framework in Figure 1 that explains how interorganizational ties and capacity can influence local government perceptions of sustainability. Collaboration with local, regional, and private sector counterparts, including the creation of interdependencies and the joint development of solutions for wicked problems, can alter the way that local governments formulate their goals and prioritize complex solutions, such as SCT. At the same time, organizations must have sufficient capacity to prioritize work on complex policy areas like sustainability. As a result, local governments are likely to perceive the potential benefits for sustainability and SCT based on a combination of their organizational capacities and their external relationships.

Methodology

Survey and Sample

The data for this research is drawn from the 2016 International City/County Management Association (ICMA) Smart Cities Survey. Partnering with the Smart Cities Council, ICMA sent the survey in 2016 to 3,423 local government officers working in U.S. cities and counties with populations of 25,000 or greater. In all, 493 officers responded to the survey questions by email or mail, giving a response rate of 14.4%. The survey consisted of twenty questions asking officers about their local governments' smart cities practices including: their commitment to SCT (priorities), their engagement with SCT (activities), their perceived benefits of using SCT, and their barriers to implementing SCT. The survey also provides insights into the extent to which local governments pursuing SCT are collaborating with other government agencies and authorities. The survey further offers information on the financial mechanisms localities are using to fund SCT efforts and projects.

The data from the ICMA survey was merged with information on demographic, economic, and population data drawn from the 2016 U.S. Census Bureau database. After eliminating records with missing data, our final sample includes responses from 234 local governments. Most survey responses were primarily from cities (83.3%), with the remaining responses coming from counties.

Dependent Variable

The dependent variable measures the importance of sustainability benefits in motivating local governments to implement or expand the use of SCT using a five-category Likert scale, with answers ranging from (1) "not important" to (5) "very important." Local governments that respond with a (4), "important" or (5) "very important" perceive greater sustainability benefits of SCT than other governments. Very few survey respondents reported perceiving sustainability as "not important (2%)" or "less important" (12%), thus we combined these responses with moderately important to create a single category to cover all governments that perceived lower levels of sustainability benefits, called "low importance." Therefore, the dependent variable is a three-level ordinal variable with the following categories: low importance, important, very important. Table 1, which presents the frequencies of the sustainability variable, shows detailed information on how the sustainability variable was operationalized.

Independent Variables

Two main categories of independent variables are used in this study. For local governments to be able to engage in sustainability efforts, they must collaborate with several internal and external partners to exchange information and build new resources. Therefore, the first category of independent variables is collaboration, measured through three distinct variables, which are: (1) local collaboration, (2) regional collaboration, and (3) public-private partnerships. Internally, cities need to work to establish strategic objectives, coordinate efforts, and implement programs. Local governments must work to gain commitments from other jurisdictions, access innovative technologies, and understand the priorities of residents (Abels, 2014). In addition, they must work with state and local partners to identify and secure resources and a supportive policy environment. We hold that local governments that are actively engaged in collaborative initiatives like these will be more likely to prioritize sustainability, especially as a part of their smart city initiatives.

The second category of independent variables is organizational capacity. As argued previously, local governments interested in pursuing sustainability programs need to foster five types of organizational capacities: internal capacity, procurement capacity, project management capability, leadership support, and financial resources, which represent important potential barriers for communities to implement smart cities programs. As shown in Table 2, the first four of these variables are each operationalized by asking survey participants to what extent each capacity represents a barrier to implement smart technologies. The questions used a five-category Likert scale, with answers ranging from (1) "not barrier" to (5) "very significant barrier." We measure financial resources as an ordinal variable that represents the annual budget for components of SCT. Local governments need to have sufficient tangible financial resources to invest in partnerships, policy change, and technologies.

In each case, these variables are measured in terms of the presence of barriers. As previously noted, organizational capacity barriers have long been acknowledged as important hindrances for the successful consideration and implementation of policies and programs (Gargan, 1981; Hackler & Saxton, 2007; Joaquin & Greitens, 2012; Long, 1990;

Table 1. Distribution of the Dependent Variable (Sustainability Benefits)

Sustainability benefits	Frequency	Percent	Cumulative
Low importance (1)	80	34.19	34.19
Important (2)	80	37.61	71.79
Very important (3)	66	28.21	100
Total	234	100	

Van Slyke, 2007). Gargan (1981) notes that “capacity is increasingly a problem because of the growing gap between local practices and the magnitude of the demands placed upon local governments” (p. 653). Sustainability initiatives place a large burden on local governments (Swann & Deslatte, 2019), and capacity barriers may present substantial hurdles to complex partnerships and policy implementation (Sandfort, 1999; Meier & O’Toole, 2006). As a result, operationalizing and measuring these variables as barriers is appropriate, though it means that we need to interpret the results of our regression analysis accordingly. Together, these variables allow us to assess the relative importance of different kinds of tangible and intangible capacity, which may influence how local governments are able to pursue sustainable, smart cities objectives.

Control Variables

There are several other factors that might influence a local government’s perceptions of sustainability and SCT. To limit the chances of omitted variable bias, we employ a range of control variables to account for these other influences. First, there is evidence that wealthier, more educated localities will be more likely to support or consider sustainability programs (Homsy & Warner, 2015; Kwon et al., 2014; Opp et al., 2013; Pierce et al., 2011; Swann & Deslatte, 2019). Therefore, we include controls for local poverty rate (measured as the proportion of people in the community below the U.S. Census Bureau’s poverty line) and the level of education (measured as the proportion of the population with a bachelor’s degree or higher). Second, government structure might influence sustainability engagement. On one hand, a local government with a council-manager structure might be more likely to pursue sustainability policies. Through specialized training and the socialization process within professional planning associations, managers might be more responsive to smart growth, sustainable development and comprehensive planning (Saha, 2009). When sustainability policies are in line with their professional norms, managers are more likely to adopt such policies (Swann & Deslatte, 2019). On the other hand, due to political constraints like community pressure, the mayor council form of government may pursue sustainability with more force than other forms of government (Saha, 2009; Swann & Deslatte, 2019). Thus, we include a dichotomous indicator of the government structure to account for whether the local government uses a council-manager form of government or not. Finally, we include dichotomous indicators to control for regional political and cultural differences across the United States (Berman, 2001).

Table 2 provides specific details about how each of these variables is operationalize, including the source question(s) for the data in the 2016 ICMA Smart Cities Survey. Descriptive statistics for each variable can be found in Table 3.

Estimation Strategy

To address the proposed hypotheses, ordinal logistic regression, a form of maximum likelihood estimation (MLE), was performed. This approach is appropriate since the dependent variable is a categorical variable with three ordered levels. The equation for the relationship between the perceived importance of sustainability benefits and independent variables is shown below:

$$Y_i = B_0 + B_1 \text{ Collaborative Governance} + B_2 \text{ Organizational Capacity} + B_3 \text{ Control Variables} + \epsilon$$

Y represents the dependent variable, perceived sustainability benefits. Vectors of variables measuring aspects of collaborative governance and organizational capacity are the main independent variables. To ascertain the appropriateness of the ordinal logistic regression, several procedures were conducted to check for the assumptions required for this type of regressions. First, multicollinearity was checked among the independent variables. The model does not suffer from multicollinearity as the mean variance inflation factor was 1.31, and the maximum VIF was 1.58.

Table 2. Variables Operationalization, Description, and Sources

Variables	Measure/description
Dependent variable Sustainability benefits	An ordinal variable with three categories in which the participants responded to the question “How important are each of the following benefits in motivating your local government to implement or expand the use of SCT?” Ordinal variable with three (1 = sustainability benefits are perceived low importance; 2 = sustainability benefits are perceived as important; 3 = sustainability benefits are perceived very important) Source. 2016 ICMA Survey
Independent variables Collaborative governance Local collaboration	A binary variable which attains a value of 1 if the community participates in collaborative efforts at county level, including peer-to-peer information exchange (city staff to city staff, within and/or across cities, etc.), and 0 otherwise. The participants responded to the question “What collaborative smart city efforts does your community participate in? and county level efforts was one of the given options. Source. 2016 ICMA Survey
Regional collaboration	A binary variable which attains a value of 1 if the community participates in collaborative efforts in state/regional system efforts or organized by regional authorities such as district], and 0 otherwise. The participants responded to the question “What collaborative smart city efforts does your community participate in? and State/regional system level efforts was one of the given options. Source. 2016 ICMA Survey
Public/private collaboration	A binary variable which attains a value of 1 if the community uses collaborative public-private partnerships to undertake smart city procurement in the community, and 0 otherwise. The participants responded to the question “Which methods are commonly used for undertaking smart city procurements in your community?” Source. 2016 ICMA Survey
Organizational capacity Internal capacity	Likert Scale with five categories treating them as a continuous variable in which the participants select “to what extent does internal capacity represent a barrier for the community to implement smart technologies?” 1 = Not a barrier 2 = Less significant barrier 3 = Moderate barrier 4 = Significant barrier 5 = Very significant barrier Source. 2016 ICMA Survey
Complexity of procurement	Likert Scale with five categories treating them as a continuous variable in which the participants select “to what extent does procurement complexity represent a barrier for the community to implement smart technologies?” 1 = Not a barrier 2 = Less significant barrier 3 = Moderate barrier 4 = Significant barrier 5 = Very significant barrier Source. 2016 ICMA Survey

(continued)

Table 2. (continued)

Variables	Measure/description
Project management capacity	Likert Scale with five categories treating them as a continuous variable in which the participants select “to what extent does project management capacity represent a barrier for the community to implement smart technologies?” An ordinal variable with five levels: 1 = Not a barrier 2 = Less significant barrier 3 = Moderate barrier 4 = Significant barrier 5 = Very significant barrier Source. 2016 ICMA Survey
Leadership support	Likert Scale with five categories treating them as a continuous variable in which the participants select “to what extent does leadership support represent a barrier for the community to implement smart technologies?” 1 = Not a barrier 2 = Less significant barrier 3 = Moderate barrier 4 = Significant barrier 5 = Very significant barrier Source. 2016 ICMA Survey
Financial capacity	An ordinal variable with five levels in which the participants select “the annual budget for components of Smart Cities Technologies (software, hardware, and consulting).” 1 = Under \$1 million 2 = From \$1 million to \$5 million 3 = Over \$5 million Source. 2016 ICMA Survey
Control variables	
Education (%)	Percent of population with bachelor's degree or higher Source. U.S. Census Bureau
Poverty (%)	The proportion of the population whose income in the past 12 months is below the poverty level Source. U.S. Census Bureau
Form of government (council-manager)	A binary variable which attains a value of 1 if the form of government is Council-manager or Council-administrator, and 0 otherwise. Source. 2016 ICMA Survey
Regions	Three binary variables created for the “region” (1 = whether local government is located in the region; 0 = it not located within the region) The regions are Midwest and Northeast regions (combined because of low frequencies), South Region, and West Region (reference group) Source. 2016 ICMA Survey

Table 3. Descriptive Statistics

Variables	Mean	Std. dev.	Min	Max
Dependent variable				
Sustainability benefits	1.94	0.79	1	3
Independent variables				
Collaborative governance				
Local collaboration	0.75	0.44	0	1
Regional collaboration	0.47	0.5	0	1
Public/private collaboration	0.43	0.5	0	1
Organizational capacity				
Internal capacity	3.6	1.07	1	5
Procurement complexity	2.77	0.93	1	5
Project management capacity	3.05	1	1	5
Financial capacity	1.44	0.61	1	3
Leadership support	2.64	1.16	1	5
Control variables				
Education (%)	32.75	16.1	7.1	74.2
Poverty (%)	15.73	7.89	2.6	49.3
Form of government (council-manager)	0.8	0.4	0	1
Midwest & northeast	0.28	0.45	0	1
South	0.36	0.48	0	1
West	0.36	0.48	0	1

Second, Brant tests of parallel regression assumption and a likelihood-ratio test of proportionality of odds are conducted. The test results were insignificant with p -value of .908 and .916 for both tests accordingly, which provide evidence that the parallel regression assumption and test of proportionality of odds have not been violated.

Third, since both our dependent variable and independent variable come from a single survey, we need to address the possibility of common source bias. The Harman single factor test, which is a common measure to detect common source bias (Harman, 1976), was run and indicated that a single factor in the Harman test was extracting 13.1 % of total variance. Since this value is far less than 50%, we conclude that there is no evidence of common method bias in our model.

Results and Discussion

Tables 4 and 5 report the results of the regression analysis, presenting the average marginal effects (AME) (Table 4) and raw coefficients and odds ratios (Table 5). To simplify explanation, we primarily discuss the average marginal effects (AME) as these are preferred for clarity of the results in these types of models (Long & Freese, 2006).

We hypothesize that collaborative relationships are associated with perceived sustainability benefits for SCT adoption. We find evidence that local governments engaging in peer-to-peer (local) partnerships are approximately 16% points less likely to perceive low sustainability benefits associated with the adoption of SCT ($p < .01$, H1a) and are 14% points more likely to perceive sustainability benefits as very important for SCT adoption ($p < .01$, H1a). We also find that local governments engaged in regional collaboration are 10% points less likely to perceive low sustainability benefits of SCT ($p < .1$, H1b) and 9% points more likely to perceive sustainability benefits of SCT as very important ($p < .05$, H1b). Regarding public-private partnerships, local governments are 9% points less likely to perceive low sustainability benefits of SCT ($p < .1$, H1c) and are 8% points more likely to perceive very important sustainability benefits ($p < .1$, H1c). Taken together, these three findings offer strong support for hypotheses H1a, H1b, and H1c: local governments engaging in collaboration are generally more likely to perceive sustainability as a primary benefit of adopting SCT.

Collaboration has been shown to have many benefits when it comes to addressing wicked problems, including helping to define problems and identify mutually beneficial solutions (Head & Alford, 2015). At the most basic level, collaboration may help make local governments aware of policy innovations, as they will hear from partners about new

Table 4. Marginal Effects Determinants on Perceived Sustainability Benefits

Variables	Low importance	Important	Very important
Collaborative governance			
Local collaboration	-0.159***	0.022	0.137***
Regional collaboration	-0.102*	0.014	0.088**
Public/private collaboration	-0.092*	0.013	0.079*
Organizational capacity			
Internal capacity	-0.046**	0.006	0.039**
Complexity in procurement	-0.048	0.007	0.042
Project management capacity	0.022	-0.003	-0.019
Financial capacity	-0.094**	0.013	0.081*
Leadership support	0.043**	-0.006	-0.037**
Control variables			
Education	-0.005***	0.001	0.005***
Poverty	-0.010**	0.001	0.009**
Form of government (council-manager)	0.141**	-0.02	-0.121**
Midwest/northeast	0.049	-0.007	-0.042
South	0.044	-0.006	-0.037

Note. Estimates are computed with margin option (mchange, atmeans centered). Plus one centered effects are reported for a given independent variable X while holding other covariates at their means.

* $p < .1$. ** $p < .05$. *** $p < .01$.

Table 5. Effects of Determinants on Perceived Sustainability Benefits

Variables	Coefficients (SE)	Odds ratio
Collaborative governance		
Local collaboration	0.74 (0.27) ***	2.09 ***
Regional collaboration	0.47 (0.26) *	1.60 *
Public/private collaboration	0.42 (0.25) *	1.53*
Capacity		
Internal capacity	0.21 (0.09)**	1.23**
Complexity in procurement	0.22 (0.15)	1.25
Project management capability	-0.10 (0.11)	0.91
Financial capacity	0.43 (0.21)**	1.54**
Leadership	-0.20 (0.09)**	0.82**
Controls		
Education (%)	0.02 (0.01)***	1.02***
Poverty (%)	0.05 (0.02)***	1.05***
Form of government (council-manager)	-0.65 (0.31)**	0.52**
Midwest northeast	-0.22 (0.36)	0.80
South	-0.20 (0.31)	0.82
/cut1	2.25 (0.71)***	9.45***
/cut2	4.09 (0.72)***	59.92***
Observations	234	234
Prob > χ^2	0.00	0.00
Pseudo R^2	0.08	0.08
Brant test	0.908	0.908

Note. Robust standard errors in parentheses.

* $p < .1$. ** $p < .05$. *** $p < .01$.

programs and their details (McGuire, 2006; Meijer and Bolívar, 2016). Over time, collaboration also contributes to learning and capacity-building, which can strengthen ties between partners (Dimitrijevska-Markoski & Nukpezah, 2022). In local governance, collaboration connects public managers, allowing information sharing and facilitating

collective actions to address shared problems (Soujaa et al., 2021). This is especially important in areas of policy innovation and high task complexity, where uncertainty, resource gaps, and information asymmetries can lead to disparities in the adoption and implementation of new programs (Hansen et al., 2020; Lee, 2011). For local governments, working with partners “plays an important role in negotiating agreements and reducing the environmental uncertainty” that surrounds the adoption of new technologies and novel modes of service delivery (Carr & Hawkins, 2013, p. 236).

SCT, which advanced technological solutions supporting a range of local initiatives, are state-of-the-art systems that present local governments with uncertain requirements, implementation challenges, and unknown outcomes. As a result, collaboration with partners may be an effective way to learn from peer experiences, facilitate effective program design, and reduce transaction costs associated with lowering uncertainty. Our results show that local governments actively engaged in collaboration perceive higher sustainability benefits from the adoption of SCT than more insular governments. This suggests that partnership plays an important role in making the connection between technological investments and sustainable development goals. The strongest effects come from local-local collaboration, indicating that mutuality and goal congruence may influence the perception of sustainability benefits. However, all types of collaboration are positively associated with perceived sustainability benefits, suggesting that governments engaged in complex partnerships are more likely to prioritize the sustainability benefits of new technology. This makes sense, as sustainability goals are most likely to be effective as part of a broader network of intergovernmental commitments, policies, and programs (Keiner & Kim, 2007). Our findings demonstrate that collaboration at the local level increases the technology-sustainability linkage, likely due to lower information costs and reduced uncertainty.

We also hold that organizational capacity is associated with positive perceptions of sustainability benefits of SCT, as capacity is needed to both recognize and adopt SCT. We hypothesized that local governments with higher internal capacity would be more likely to perceive sustainability benefits of SCT. Our findings indicate that local governments that do not have internal capacity barriers are 5% points less likely to see low levels of sustainability benefits of SCT ($p < .05$), and 4% points more likely to perceive very important sustainability benefits of SCT ($p < .05$, H2a). Consistent with prior research on sustainability (e.g., Deslatte et al., 2021; Hawkins et al., 2021), our findings also note that local governments with high internal capacity are more likely to prioritize sustainability benefits. Implementing new programs puts a toll on the existing resources and increased capacity is needed (Koerner & Johnston, 2021).

We hypothesize that local governments with fewer leadership capacity problems would be more likely to perceive sustainability benefits of SCT (H2d). Our findings indicate that governments with lower leadership barriers are likely to perceive low sustainability benefits associated with SCT (by about 4% points), while governments with higher leadership barriers are less likely to see very important sustainability benefits (by about 4% points). This is indicative of the importance of leaders as champions: in local governments where leaders present significant barriers to the adoption of SCT, sustainability benefits are perceived as less important. Additionally, consistent with the literature on organizational change (e.g., Fernandez & Rainey, 2017; Gilley et al., 2009), our results show that local governments that have leadership support have a higher probability of prioritizing sustainability benefits of SCT.

We hypothesize that governments with fewer financial barriers would be more likely to perceive sustainability benefits of SCT. Our results show that local financial capacity is negatively associated with a perception of low sustainability benefits of SCT (about 9% points lower) and positively associated with a perception that sustainability benefits are very important to SCT (by about 8% points). This suggests that when a local government has sufficient financial resources, it prioritizes sustainability elements of SCT (H2e). Sustainability initiatives like smart cities are highly innovative (Appio et al., 2019) and impose significant financial burdens on organizations (Hawkins et al., 2016; Opp et al., 2014). Recent studies indicate that access to financial resources is a necessary condition for organizational change (Fernandez & Rainey, 2017), and that fiscal constraints have been frequently associated with lower levels of innovation (Schick, 1988; Walker & Chaiken, 1982). Consistent with these perspectives, we find that local governments with the necessary financial resources to consider and implement innovative policy changes are more likely to prioritize sustainability benefits.

Contrary to our hypotheses, procurement capacity and project management capacity do not influence perceptions of sustainability benefits of SCTs, indicating that they may not be seen as barriers for local governments (H2b and H2c). There are a range of possible explanations for these findings. It is possible that local governments possess sufficient procurement and project management capacity, and accordingly do not view or experience these areas as weaknesses (Brown & Potoski, 2003). Yet, this explanation is not consistent with more recent studies that indicate that procurement and project management skills are among the most needed in government, particularly at the local level (Atkinson

& Sapat, 2013; Bromberg & Manahoran, 2015; Joaquin & Greitens, 2012). It might also be the case that for many governments that have long been dedicated to direct delivery of public services, procurement and project management expertise is forgotten or neglected (Hou et al., 2003; Milward & Provan, 2000; Snider & Rendon, 2012; Thai & Grimm, 2000; Trammell et al., 2020). If these skills are under-acknowledged, it is likely that survey respondents may also be either unaware of their importance or unwittingly overestimate the capacities of their own local governments. It is also possible that survey respondents might think about sustainability in terms of other aspects of public agencies instead of procurement. This offers fertile ground for future studies, as parsing the relative importance of collaboration, organizational capacity, and other factors would improve our understanding of how governments and public officials think about and prioritize sustainability.

A few of our control variables are also worth noting, as they are related to perceptions of SCT sustainability benefits. Education and poverty levels conform to expectations: richer, more highly educated localities are more likely to perceive higher sustainability benefits of smart cities programs. From a capacity perspective, such communities may have more organizational capacity to consider sustainability benefits, may ascribe greater value to sustainable public policies, and are more likely to have leadership interest in promoting sustainability (Bowman & Kearney, 2016; Krause, 2011). There is also evidence that wealthier, educated communities are more likely to engage in collaboration, as they actively seek both information and increased regional power (Agranoff & McGuire, 2003). Further, we find that local governments with council-manager or council-administrator forms of government are also more likely to perceive sustainability benefits of SCT compared to other forms of government. Administrative professionalization leads to increased perceptions of benefits, perhaps evidence that training and protection from politics (to some extent) enable local governments to make the connection between innovative technologies and sustainability.

Conclusion

Local governments are increasingly seeking tools to achieve sustainability benefits (Ji & Darnall, 2018), and one such innovative solution is the use of SCT (Lim et al., 2019). One way local governments learn about new tools is through collaboration with partners, including other governments and private firms. This is especially true in the 21st century, as many of the problems we face span boundaries and require the involvement of many organizations. However, organizational capacity is needed to adopt and implement innovations (Andrews & Boyne, 2010) such as SCT. Organizations can rely on internal capacity or seek resources from the external environment as they strive to identify new policy solutions.

This research investigates how collaboration and organizational capacity influence the perception of sustainability benefits associated with US local government adoption of SCT. Using ICMA survey data from 243 localities, we demonstrate that local governments involved in collaborative partnerships with local, regional, and private-sector organizations are more likely to list sustainability benefits as a primary motivation for pursuing the adoption or implementation of SCT. Traditionally viewed as efficiency initiatives designed to improve the responsiveness and integration of local government systems (Ruhlandt, 2018), smart cities efforts may not be obviously related to sustainability. Yet, for cities committed to boundary spanning, cross-sectoral partnerships, sustainability is seen as a critical reason for pursuing SCT. That is, engaging in collaborative governance leads local governments to (a) recognize the ways that new technologies can help promote sustainability objectives and (b) identify sustainability as a rationale for engaging in the complex work of integrating policy and management systems. This makes sense, as addressing wicked problems, like sustainability, often requires partnerships and collaboration to share information, engage in productive sensemaking, develop potential solutions, and share needed resources (Agranoff & McGuire, 2003; Angus-Leppan et al., 2010; Head & Alford, 2015).

For scholars, our findings suggest that more research is needed into the ways that collaboration can influence organizational goals and priorities. If boundary spanning activities can influence perceptions of sustainability benefits, it is also likely that partnerships can affect perceptions of other policy and management priorities. If, as advocacy coalition scholars assert, organizations enter partnerships with malleable beliefs and a willingness to change (Jenkins-Smith, Nohrstedt et al., 2018), it is reasonable to expect that collaboration may affect organizational management, including the conceptualization of key issues, categorization and ranking of objectives, and allocation of resources. Additional research is needed to understand how collaboration may influence how organizations perceive and respond to complex problems.

We also show that elements of organizational capacity are needed for local governments to perceive sustainability benefits. Organizations with fewer barriers to internal capacity, financial resources, and leadership support are more

likely to perceive sustainability objectives as a key motivation for pursuing SCT. Joaquin and Greitens (2021) note that capacity provides “the floor, from which any reduction or disinvestment can produce breakdowns or failures” (p. 48). Agencies need a range of capacities to engage in collaborative governance (Emerson et al., 2012), work to address wicked problems (Weber & Khademian, 2008), and to adopt complex technologies (Bozeman, 2000; Karo & Kattel, 2018). Consistent with these studies, our findings show that local governments with fewer capacity barriers are more likely to pursue sustainable technologies. For practitioners, investing in internal capacity and seeking financial resources may create favorable conditions for the improved perception and increased adoption of smart cities approaches.

Our study is not without limitations. One limitation of the study is the measurement of the variables using a perception-based instrument. Though surveys have long been central to empirical analysis in public administration (Pandey, 2017), future studies could collect objective measures for each of the items and assess whether the results are consistent. Scholars have recently noted an exaggeration in criticism toward common source bias (George & Pandey, 2017) because data for both the dependent and independent variables are drawn from the same data source. We conducted a Harman single factor test, and the results show no evidence of common method bias in our model. Our study is also cross-sectional, and accordingly cannot account for changes over time. Future research should investigate temporal elements of the linkages between sustainability, collaboration, and organizational capacity. Similarly, future studies could examine the extent to which possible sustainability benefits are achieved (or recognized) after the adoption of SCT.

Finally, though we establish an initial link here between collaboration and sustainability, more research is needed to clarify the causal relationship between these two variables. Though we find evidence that collaborative governance is associated with a higher motivation to pursue sustainability benefits, it is also possible that local governments interested in sustainability may be more likely to collaborate to address this wicked problem. For us, this is not necessarily a problem, as regardless of the causal path, we offer new evidence that local government collaboration and the pursuit of sustainability are linked. Yet, future scholarship should further explore the directionality of this correlation: we believe that it is likely to work both ways in practice, as some local governments (or their collaborative partners) will intentionally engage in collaboration to help advance their sustainability goals, while others may learn of (or reprioritize) sustainability programs and policies based on interactions with their collaborators.

As the importance of local governments in sustainability initiatives grows, it is necessary to understand what contributes to the motivation to pursue sustainability objectives. In this study, we show that collaboration may lead local governments to adopt smart, sustainable programs, and that management and financial capacity are also critical. Building on other studies of collaboration, capacity, and sustainability, we offer further evidence that to solve the wicked problems of the 21st century, it is necessary to work together and to allocate the necessary financial and human capital to public organizations.

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Note

1. While the literature generally pinpoints a positive relationship between smart cities and sustainability (Lim et al., 2019), there are also studies that do not identify such results when focusing on the environmental aspect of sustainability (Yigitcanlar et al., 2018).

References

- Abels, M. (2014). Strategic alignment for the new normal: Collaboration, sustainability, and deliberation in local government across boundaries. *State and Local Government Review*, 46(3), 211–218. <https://doi.org/10.1177/0160323X14551179>
- Abutabenjeh, S., Dimand, A. M., & Tao, J. (2023). Determinants of the success of cooperative public procurement. *Public Performance & Management Review*, 46(2), 391–417.
- Abutabenjeh, S., Nukpezah, J. A., & Azhar, A. (2022). Do smart cities technologies contribute to local economic development? *Economic Development Quarterly*, 36(1), 3–16. <https://doi.org/10.1177/0891242421105359>

- Acuto, M. (2013). City leadership in global governance. *Global Governance*, 19, 481–498.
- Agranoff, R. (2017). *Crossing boundaries for intergovernmental management*. Georgetown University Press.
- Agranoff, R., & McGuire, M. (2003). *Collaborative public management: New strategies for local governments*. Georgetown University Press.
- Alfaro Navarro, J. L., López Ruiz, V. R., & Nevado Peña, D. (2017). The effect of ICT use and capability on knowledge-based cities. *Cities*, 60, 272–280. <https://doi.org/10.1016/j.cities.2016.09.010>
- Andrews, R., & Boyne, G. A. (2010). Capacity, leadership, and organizational performance: Testing the black box model of public management. *Public Administration Review*, 70(3), 443–454. <https://doi.org/10.1111/j.1540-6210.2010.02158.x>
- Angelidou, M. (2015). Smart cities: A conjuncture of four forces. *Cities*, 47, 95–106. <https://doi.org/10.1016/j.cities.2015.05.004>
- Angelidou, M. (2017). The role of smart city characteristics in the plans of fifteen cities. *Journal of Urban Technology*, 24(4), 3–28. <https://doi.org/10.1080/10630732.2017.1348880>
- Angus-Leppan, T., Benn, S., & Young, L. (2010). A sensemaking approach to trade-offs and synergies between human and ecological elements of corporate sustainability. *Business Strategy and the Environment*, 19(4), 230–244. <https://doi.org/10.1002/bse.675>
- Ansell, C., & Gash, A. (2007). Collaborative governance in theory and practice. *Journal of Public Administration Research and Theory*, 18(4), 543–571. <https://doi.org/10.1093/jopart/mum032>
- Appio, F. P., Lima, M., & Paroutis, S. (2019). Understanding smart cities: Innovation ecosystems, technological advancements, and societal challenges. *Technological Forecasting and Social Change*, 142, 1–14. <https://doi.org/10.1016/j.techfore.2018.12.018>
- Aral, E. (2020). Why do cities adopt smart technologies? Contingency theory and evidence from the United States. *Cities*, 106, 102873. <https://doi.org/10.1016/j.cities.2020.102873>
- Atkinson, C. L., & Sapat, A. K. (2013). Hurricane Wilma and long-term business recovery in disasters: The role of local government procurement and economic development. *Journal of Homeland Security and Emergency Management*, 11(1), 169–192. <https://doi.org/10.1515/jhsem-2013-0002>
- Awasthi, R., Nagarajan, M., & Deininger, K. W. (2021). Property taxation in India: Issues impacting revenue performance and suggestions for reform. *Land Use Policy*, 110, 104539. <https://doi.org/10.1016/j.landusepol.2020.104539>
- Bakıcı, T., Almirall, E., & Wareham, J. (2013). A smart city initiative: The case of Barcelona. *Journal of the Knowledge Economy*, 4(2), 135–148. <https://doi.org/10.1007/s13132-012-0084-9>
- Bankins, S., Denness, B., Kriz, A., & Molloy, C. (2017). Innovation agents in the public sector: Applying champion and promoter theory to explore innovation in the Australian public service: Innovation agents in the public sector. *Australian Journal of Public Administration*, 76(1), 122–137. <https://doi.org/10.1111/1467-8500.12198>
- Batty, M. (2020). Defining smart cities: High and low frequency cities, big data and urban theory. In K. Kitchin, S. Marvin, & A. Perng (Eds.), *The Routledge companion to smart cities* (1st ed., pp. 10–24). Routledge. <https://doi.org/10.4324/9781315178387>
- Bell, S., & Morse, S. (2003). *Measuring sustainability: Learning by doing*. Earthscan Publications Ltd.
- Benavides, A. D., & Nukpezah, J. A. (2020). How local governments are caring for the homeless during the COVID-19 pandemic. *The American Review of Public Administration*, 50(6–7), 650–657. <https://doi.org/10.1177/0275074020942062>
- Berman, S. (2001). Ideas, norms, and culture in political analysis. *Comparative Politics*, 33(2), 231–250.
- Bertelli, A. M., & Smith, C. R. (2010). Relational contracting and network management. *Journal of Public Administration Research and Theory*, 20(Suppl1), i21–i40. <https://doi.org/10.1093/jopart/mup033>
- Bianchi, C., Nasi, G., & Rivenbark, W. C. (2021). Implementing collaborative governance: Models, experiences, and challenges. *Public Management Review*, 23(11), 1581–1589. <https://doi.org/10.1080/14719037.2021.1878777>
- Bjorvatn, T., & Wald, A. (2018). Project complexity and team-level absorptive capacity as drivers of project management performance. *International Journal of Project Management*, 36(6), 876–888. <https://doi.org/10.1016/j.ijproman.2018.05.003>
- Bowman, A. O. M., & Kearney, R. C. (2016). *State and local government*. Cengage Learning.
- Bozeman, B. (2000). Technology transfer and public policy: A review of research and theory. *Research Policy*, 29(4–5), 627–655. [https://doi.org/10.1016/S0048-7333\(99\)00093-1](https://doi.org/10.1016/S0048-7333(99)00093-1)
- Brink, E., & Wamsler, C. (2018). Collaborative governance for climate change adaptation: Mapping citizen–municipality interactions. *Environmental Policy and Governance*, 28(2), 82–97. <https://doi.org/10.1002/eet.1795>
- Bromberg, D., & Manoharan, A. (2015). E-procurement implementation in the United States: Understanding progress in local government. *Public Administration Quarterly*, 39, 360–392.
- Brown, T. L., & Potoski, M. (2003). Contract-management capacity in municipal and county governments. *Public Administration Review*, 63(2), 153–164. <https://doi.org/10.1111/1540-6210.00276>
- Brown, T. L., Potoski, M., & Van Slyke, D. M. (2013). *Complex contracting*. Cambridge University Press.
- Brown, T. L., Potoski, M., & Van Slyke, D. M. (2006). Managing public service contracts: Aligning values, institutions, and markets. *Public Administration Review*, 66(3), 323–331. <https://doi.org/10.1111/j.1540-6210.2006.00590.x>
- Brunjes, B. M. (2020). Competition and federal contractor performance. *Journal of Public Administration Research and Theory*, 30(2), 202–219. <https://doi.org/10.1093/jopart/muz027>
- Burke, R. (2013). Project management: Planning and control techniques. *John Wiley & Sons*.
- Burns, R., & Andrucki, M. (2021). Smart cities: Who cares?. *Environment and Planning A: Economy and Space*, 53(1), 12–30. <https://doi.org/10.1177/0308518X2094151>

- Caragliu, A., & Del Bo, C. F. (2019). Smart innovative cities: The impact of Smart City policies on urban innovation. *Technological Forecasting and Social Change*, 142, 373–383.
- Carr, J. B., & Hawkins, C. V. (2013). The costs of cooperation: What the research tells us about managing the risks of service collaborations in the US. *State and Local Government Review*, 45(4), 224–239.
- Christensen, R. K., & Gazley, B. (2008). Capacity for public administration: Analysis of meaning and measurement. *Public Administration and Development: The International Journal of Management Research and Practice*, 28(4), 265–279. <https://doi.org/10.1002/pad.500>
- Christensen, T., & Lægreid, P. (2011). Complexity and hybrid public administration—theoretical and empirical challenges. *Public Organization Review*, 11(4), 407–423. <https://doi.org/10.1007/s11115-010-0141-4>
- Clever, S., Crago, T., Polka, A., Al-Jaroodi, J., & Mohamed, N. (2018). Ethical analyses of smart city applications. *Urban Science*, 2(4), 96. <https://doi.org/10.3390/urbansci2040096>
- Cutter, S. L., Barnes, L., Berry, M., Burton, C., Evans, E., Tate, E., & Webb, J. (2008). A place-based model for understanding community resilience to natural disasters. *Global Environmental Change*, 18(4), 598–606. <https://doi.org/10.1016/j.gloenvcha.2008.07.013>
- Dameri, R. P., & Cocchia, A. (2013, December). Smart city and digital city: Twenty years of terminology evolution [Conference session]. In *X Conference of the Italian Chapter of AIS, ITAIS* (pp. 1–8). Milan, Italy.
- Dameri, R. P., & Ricciardi, F. (2015). Smart city intellectual capital: An emerging view of territorial systems innovation management. *Journal of Intellectual Capital*, 16(4), 860–887. <https://doi.org/10.1108/JIC-02-2015-0018>
- Daviter, F. (2017). Coping, taming or solving: Alternative approaches to the governance of wicked problems. *Policy Studies*, 38(6), 571–588. <https://doi.org/10.1080/01442872.2017.1384543>
- Deslatte, A., & Stokan, E. (2020). Sustainability synergies or silos? The opportunity costs of local government organizational capabilities. *Public Administration Review*, 80(6), 1024–1034. <https://doi.org/10.1111/puar.13237>
- Deslatte, A., Swann, W. L., & Feiock, R. C. (2021). Performance, satisfaction, or loss aversion? A Meso–Micro assessment of local commitments to sustainability programs. *Journal of Public Administration Research and Theory*, 31(1), 201–217. <https://doi.org/10.1093/jopart/muaa021>
- Diallo, A. B., & Reid, M. F. (2021). Sustaining cross-sector collaborations: Managing conflicts in large-scale urban event collaborations. *Public Administration Quarterly*, 45(4). <https://doi.org/10.37808/paq.45.4.5>
- Dimitrijevska-Markoski, T., & Nukpezah, J. A. (2022). Determinants of network effectiveness: Evidence from a benchmarking consortium. *International Journal of Public Sector Management*, 35(1), 61–75.
- Dyer, J. H. (1997). Effective interim collaboration: How firms minimize transaction costs and maximise transaction value. *Strategic Management Journal*, 18(7), 535–556.
- Emerson, K., Nabatchi, T., & Balogh, S. (2012). An integrative framework for collaborative governance. *Journal of Public Administration Research and Theory*, 22(1), 1–29. <https://doi.org/10.1093/jopart/mur011>
- Eriksson, E., Andersson, T., Hellström, A., Gadolin, C., & Lifvergren, S. (2020). Collaborative public management: Coordinated value propositions among public service organizations. *Public Management Review*, 22(6), 791–812.
- Estoque, R. C., & Murayama, Y. (2014). Measuring sustainability based upon various perspectives: A case study of a hill station in southeast Asia. *AMBIO*, 43(7), 943–956. <https://doi.org/10.1007/s13280-014-0498-7>
- Fernandez, S. (2004). Developing and testing an integrative framework of public sector leadership: Evidence from the public education arena. *Journal of Public Administration Research and Theory*, 15(2), 197–217. <https://doi.org/10.1093/jopart/mui014>
- Fernandez, S., & Rainey, H. G. (2017). Managing successful organizational change in the public sector. In R. F. Durant & J. R. S. Durant (Eds.), *Debating public administration: Management challenges, choices, and opportunities* (pp. 7–26). Routledge. <https://doi.org/10.4324/9781315095097>
- Fiorino, D. J. (2010). Sustainability as a conceptual focus for public administration. *Public Administration Review*, 70, s78–s88. <https://doi.org/10.1111/j.1540-6210.2010.02249.x>
- Fiorino, D. J., & Weted, C. A. (2020). Environmental federalism in a polarized era. *State and Local Government Review*, 52(2), 138–151. <https://doi.org/10.1177/0160323X20986225>
- Forrer, J., & Kee, J. E. (2004). Public servants as contract managers? *Public Contract Law Journal*, 33(2), 361–367.
- Gabrys, J. (2014). Programming environments: Environmentality and citizen sensing in the smart city. *Environment and Planning D: Society and Space*, 32(1), 30–48. <https://doi.org/10.1068/d1681>
- Gareis, R. (2010). Changes of organizations by projects. *International Journal of Project Management*, 28(4), 314–327. <https://doi.org/10.1016/j.ijproman.2010.01.002>
- Gargan, J. J. (1981). Consideration of local government capacity. *Public Administration Review*, 41(6), 649. <https://doi.org/10.2307/975741>
- George, B., & Pandey, S. K. (2017). We know the Yin—But where is the Yang? Toward a balanced approach on common source bias in public administration scholarship. *Review of Public Personnel Administration*, 37(2), 245–270. <https://doi.org/10.1177/0734371X17698189>
- Gilley, A., Gilley, J. W., & McMillan, H. S. (2009). Organizational change: Motivation, communication, and leadership effectiveness. *Performance Improvement Quarterly*, 21(4), 75–94. <https://doi.org/10.1002/piq.20039>

- Hackler, D., & Saxton, G. D. (2007). The strategic use of information technology by nonprofit organizations: Increasing capacity and untapped potential. *Public Administration Review*, 67(3), 474–487. <https://doi.org/10.1111/j.1540-6210.2007.00730.x>
- Hansen, K., Mullin, M., & Riggs, E. K. (2020). Collaboration risk and the choice to consolidate local government services. *Perspectives on Public Management and Governance*, 3(3), 223–238.
- Harman, H. H. (1976). *Modern factor analysis*. University of Chicago press.
- Hartley, J., Sørensen, E., & Torfing, J. (2013). Collaborative innovation: A viable alternative to market competition and organizational entrepreneurship. *Public Administration Review*, 73(6), 821–830. <https://doi.org/10.1111/puar.12136>
- Hawkins, C. V., Krause, R. M., & Deslatte, A. (2021). Staff support and administrative capacity in strategic planning for local sustainability. *Public Management Review*, 25, 879–900. <https://doi.org/10.1080/14719037.2021.1999667>.
- Hawkins, C. V., Kwon, S.-W., & Bae, J. (2016). Balance between local economic development and environmental sustainability: A multilevel governance perspective. *International Journal of Public Administration*, 39(11), 803–811. <https://doi.org/10.1080/01900692.2015.1035787>
- Head, B. W., & Alford, J. (2015). Wicked problems: Implications for public policy and management. *Administration & Society*, 47(6), 711–739. <https://doi.org/10.1177/0095399713481601>
- Höjer, M., & Wangel, J. (2015). Smart sustainable cities: Definition and challenges. In L. M. Hilty & B. Aebischer (Eds.), *ICT innovations for sustainability* (pp. 333–349). Springer International Publishing.
- Hollands, R. G. (2015). Critical interventions into the corporate smart city. *Cambridge Journal of Regions, Economy and Society*, 8(1), 61–77. <https://doi.org/10.1093/cjres/rsu011>
- Hölscher, K., Frantzeskaki, N., McPhearson, T., & Loorbach, D. (2019). Tales of transforming cities: Transformative climate governance capacities in New York City, US and Rotterdam, Netherlands. *Journal of Environmental Management*, 231, 843–857. <https://doi.org/10.1016/j.jenvman.2018.10.043>
- Homsy, G. C., & Warner, M. E. (2015). Cities and sustainability: Polycentric action and multilevel governance. *Urban Affairs Review*, 51(1), 46–73. <https://doi.org/10.1177/1078087414530545>
- Homsy, G. C., Liao, L., & Warner, M. E. (2019). Sustainability and disaster planning: What are the connections? *Rural Sociology*, 84(3), 516–540. <https://doi.org/10.1111/ruso.12262>
- Hou, Y., Moynihan, D. P., & Ingraham, P. W. (2003). Capacity, management, and performance: Exploring the links. *The American Review of Public Administration*, 33(3), 295–315. <https://doi.org/10.1177/0275074003251651>
- ICMA. (2022). *Resilient and sustainable communities*. <https://icma.org/topics/resilient-and-sustainable-communities#:~:text=ICMA%20views%20sustainability%20as%20a,create%20thriving%20and%20livable%20communities>.
- Ingraham, P. W., Joyce, P. G., & Donahue, A. K. (2003). *Government performance: Why management matters*. Taylor & Francis.
- Jenkins-Smith, H. C., Nohrstedt, D., Weible, C. M., & Ingold, K. (2018). The advocacy coalition framework: An overview of the research program. In C. M. Weible & P. A. Sabatier (Eds.), *Theories of the Policy Process* (pp. 135–171). Routledge.
- Ji, H., & Darnall, N. (2018). All are not created equal: Assessing local governments' strategic approaches towards sustainability. *Public Management Review*, 20(1), 154–175. <https://doi.org/10.1080/14719037.2017.1293147>.
- Joaquin, M. E., & Greitens, T. J. (2012). Contract management capacity breakdown? An analysis of U.S. local governments. *Public Administration Review*, 72(6), 807–816. <https://doi.org/10.1111/j.1540-6210.2012.02587.x>
- Joaquin, M. E., & Greitens, T. J. (2021). *American administrative capacity*. Springer Books.
- Johnston, J., & Kouzmin, A. (2010). Addressing governance, accountability and performance monitoring issues in partnerships: Can 'infrastructure Australia' provide a strategic response? *Public Administration Quarterly*, 34, 513–551.
- Jukić, T., Pevcin, P., Benčina, J., Dečman, M., & Vrbek, S. (2019). Collaborative innovation in public administration: Theoretical background and research trends of co-production and co-creation. *Administrative Sciences*, 9(4), 90.
- Karo, E., & Kattel, R. (2018). Innovation and the state: Towards an evolutionary theory of policy capacity. In X. Wu, M. Howlett, & M. Ramesh (Eds.), *Policy Capacity and Governance* (pp. 123–150). Palgrave Macmillan.
- Keiner, M., & Kim, A. (2007). Transnational city networks for sustainability. *European Planning Studies*, 15(10), 1369–1395.
- Kerzner, H. (2017). *Project management: A systems approach to planning, scheduling, and controlling*. John Wiley & Sons.
- Kim, J. (2018). Collaborative leadership and financial sustainability in local government. *Local Government Studies*, 44(6), 874–893.
- Kim, Y., Aldag, A. M., & Warner, M. E. (2020). A balanced view: US local government managers' perceptions of fiscal stress. *Public Administration Quarterly*, 44(2), 172–200. <https://doi.org/10.37808/paq.44.2.2>
- Klein, J. T. (2020). Sustainability and collaboration: Cross disciplinary and cross-sector horizons. *Sustainability*, 12(4), 1515.
- Klijn, E. H., & Koppenjan, J. F. (2000). Public management and policy networks: Foundations of a network approach to governance. *Public Management and International Journal of Research and Theory*, 2(2), 135–158.
- Koerner, A. A., & Johnston, J. M. (2021). Linking organizational capacity and performance: The case of probation and Medicaid reform in California counties. *Journal of Public Administration Research and Theory*, 32(4), 702–716. <https://doi.org/10.1093/jopart/muab047>
- Komninos, N., Pallot, M., & Schaffers, H. (2013). Special issue on smart cities and the future internet in Europe. *Journal of the Knowledge Economy*, 4, 119–134. <https://doi.org/10.1007/s13132-012-0083-x>
- Krause, R. M. (2011). Policy innovation, intergovernmental relations, and the adoption of climate protection initiatives by US cities. *Journal of Urban Affairs*, 33(1), 45–60.

- Krause, R. M., Feiock, R. C., & Hawkins, C. V. (2016). The administrative organization of sustainability within local government. *Journal of Public Administration Research and Theory*, 26(1), 113–127.
- Krause, R. M., Hawkins, C. V., & Park, A. Y. (2023). How has the COVID crisis impacted local governments' sustainability efforts? An examination of initial effects. *State and Local Government Review*, 55(1), 27–40.
- Kumar, H., Singh, M. K., Gupta, M. P., & Madaan, J. (2020). Moving towards smart cities: Solutions that lead to the Smart City Transformation Framework. *Technological Forecasting and Social Change*, 153, 119281. <https://doi.org/10.1016/j.techfore.2018.04.024>
- Kushner, R. J., & Poole, P. P. (1996). Exploring structure-effectiveness relationships in nonprofit arts organizations. *Nonprofit Management and Leadership*, 7(2), 119–136. <https://doi.org/10.1002/nml.4130070203>
- Kwon, M., Jang, H. S., & Feiock, R. C. (2014). Climate protection and energy sustainability policy in California cities: What have we learned? *Journal of Urban Affairs*, 36(5), 905–924. <https://doi.org/10.1111/juaf.12094>
- Lam, P. T., & Yang, W. (2020). Factors influencing the consideration of Public-Private Partnerships (PPP) for smart city projects: Evidence from Hong Kong. *Cities*, 99, 102606. <https://doi.org/10.1016/j.cities.2020.102606>
- Larsson, J., & Larsson, L. (2020). Integration, application and importance of collaboration in sustainable project management. *Sustainability*, 12(2), 585. <https://doi.org/10.3390/su12020585>
- Laurian, L., Walker, M., & Crawford, J. (2017). Implementing environmental sustainability in local government: The impacts of framing, agency culture, and structure in US cities and counties. *International Journal of Public Administration*, 40(3), 270–283. <https://doi.org/10.1080/01900692.2015.1107738>
- Lee, Y. (2011). Economic development networks among local governments: The structure of collaboration networks in the Tampa Bay Metropolitan Area. *International Review of Public Administration*, 16(1), 113–134.
- Lehr, T. (2018). Smart cities: Vision on-the-ground. In S. McClellan, J. A. Jimenze, & G. Koutitas (Eds.), *Smart cities* (pp. 3–15). Springer.
- Leuenberger, D. Z., & Bartle, J. R. (2014). *Sustainable development for public administration*. Routledge.
- Liao, L., Warner, M. E., & Homsy, G. C. (2019). Sustainability's forgotten third E: What influences local government actions on social equity? *Local Environment*, 24(12), 1197–1208. <https://doi.org/10.1080/13549839.2019.1683725>
- Lim, W. M., & Rasul, T. (2022). Customer engagement and social media: Revisiting the past to inform the future. *Journal of Business Research*, 148, 325–342.
- Lim, Y., Edelenbos, J., & Gianoli, A. (2019). Identifying the results of smart city development: Findings from systematic literature review. *Cities*, 95, 102397. <https://doi.org/10.1016/j.cities.2019.102397>
- Lombardi, P., Giordano, S., Farouh, H., & Yousef, W. (2012). Modelling the smart city performance. *Innovation: The European Journal of Social Science Research*, 25(2), 137–149. <https://doi.org/10.1080/13511610.2012.660325>
- Long, J. S., & Freese, J. (2006). *Regression models for categorical dependent variables using Stata* (Vol. 7). Stata Press.
- Long, N. E. (1990). Conceptual notes on the public interest for public administration and policy analysts. *Administration & Society*, 22(2), 170–181. <https://doi.org/10.1177/009539979002200202>
- Lubell, M., & Lippert, L. (2011). Integrated regional water management: A study of collaboration or water politics-as-usual in California, USA. *International Review of Administrative Sciences*, 77(1), 76–100. <https://doi.org/10.1177/0020852310388367>
- Malatesta, D., & Smith, C. R. (2014). Lessons from resource dependence theory for contemporary public and nonprofit management. *Public Administration Review*, 74(1), 14–25. <https://doi.org/10.1111/puar.12181>
- Martens, M. L., & Carvalho, M. M. (2017). Key factors of sustainability in project management context: A survey exploring the project managers' perspective. *International Journal of Project Management*, 35(6), 1084–1102. <https://doi.org/10.1016/j.ijproman.2016.04.004>
- McGuire, M. (2006). Collaborative public management: Assessing what we know and how we know it. *Public Administration Review*, 66, 33–43.
- McGuire, M., & Silvia, C. (2010). The effect of problem severity, managerial and organizational capacity, and agency structure on intergovernmental collaboration: Evidence from local emergency management. *Public Administration Review*, 70(2), 279–288. <https://doi.org/10.1111/j.1540-6210.2010.02134.x>
- Meier, K. J., & O'Toole, L. J. Jr (2006). *Bureaucracy in a democratic state: A governance perspective*. JHU Press.
- Meijer, A., & Bolívar, M. P. R. (2016). Governing the smart city: A review of the literature on smart urban governance. *International Review of Administrative Sciences*, 82(2), 392–408.
- Milward, H. B., & Provan, K. G. (2000). Governing the hollow state. *Journal of Public Administration Research and Theory*, 10(2), 359–380. <https://doi.org/10.1093/oxfordjournals.jpart.a024273>
- Moldavanova, A. (2016). Two narratives of intergenerational sustainability: A framework for sustainable thinking. *The American Review of Public Administration*, 46(5), 526–545. <https://doi.org/10.1177/0275074014565390>
- Neirotti, P., De Marco, A., Cagliano, A. C., Mangano, G., & Scorrano, F. (2014). Current trends in smart city initiatives: Some stylised facts. *Cities*, 38, 25–36. <https://doi.org/10.1016/j.cities.2013.12.010>
- Nelson, K. L., & Svara, J. H. (2015). The roles of local government managers in theory and practice: A centennial perspective. *Public Administration Review*, 75(1), 49–61. <https://doi.org/10.1111/puar.12296>

- Nowell, B., Izod, A. M., Ngaruiya, K. M., & Boyd, N. M. (2016). Public service motivation and sense of community responsibility: Comparing two motivational constructs in understanding leadership within community collaboratives. *Journal of Public Administration Research and Theory*, 26(4), 663–676. <https://doi.org/10.1093/jopart/muv048>
- Nukpezah, J., Abutabenjeh, S., & Azhar, A. (2022). Do smart cities technologies contribute to revenue performance? Evidence from US local governments. *Public Performance & Management Review*, 45(5), 1155–1180. <https://doi.org/10.1080/15309576.2022.2076248>
- Okafor, C. C., Aigbavboa, C., & Thwala, W. D. (2022). A bibliometric evaluation and critical review of the smart city concept—making a case for social equity. *Journal of Science and Technology Policy Management*, 4, 487–510. <https://doi.org/10.1108/JSTPM-06-2020-0098>
- Opp, S. M., & Saunders, K. L. (2013). Pillar talk: Local sustainability initiatives and policies in the United States—Finding evidence of the “three E’s”: Economic development, environmental protection, and social equity. *Urban Affairs Review*, 49(5), 678–717. <https://doi.org/10.1177/1078087412469344>
- Opp, S. M., Osgood, J. L., & Rugeley, C. R. (2014). Explaining the adoption and implementation of local environmental policies in the United States. *Journal of Urban Affairs*, 36(5), 854–875. <https://doi.org/10.1111/juaf.12072>
- Overmans, J. F. A., & Noordegraaf, M. (2014). Managing austerity: Rhetorical and real responses to fiscal stress in local government. *Public Money & Management*, 34(2), 99–106. <https://doi.org/10.1080/09540962.2014.887517>
- Pandey, S. K. (2017). Theory and method in public administration. *Review of Public Personnel Administration*, 37(2), 131–138. <https://doi.org/10.1177/0734371X1770703>
- Park, A. Y. S., Krause, R. M., & Hawkins, C. V. (2021). Institutional mechanisms for local sustainability collaboration: Assessing the duality of formal and informal mechanisms in promoting collaborative processes. *Journal of Public Administration Research and Theory*, 31(2), 434–450. <https://doi.org/10.1093/jopart/muaa036>
- Pettibone, L. (2016). *Governing urban sustainability: Comparing cities in the USA and Germany*. Ashgate.
- Pfeffer, J., & Salancik, G. R. (1978). *The external control of organizations: A resource dependence perspective*. Stanford University Press.
- Pianezzi, D., Mori, Y., & Uddin, S. (2021). Public–private partnership in a smart city: A curious case in Japan. *International Review of Administrative Sciences*, 89(3), 632–647. <https://doi.org/10.1177/002085232110518>
- Pierce, J. C., Budd, W. W., & Lovrich, N. P. Jr. (2011). Resilience and sustainability in US urban areas. *Environmental Politics*, 20(4), 566–584. <https://doi.org/10.1080/09644016.2011.589580>
- Pihlajamaa, M., & Merisalo, M. (2021). Organizing innovation contests for public procurement of innovation—a case study of smart city hackathons in Tampere, Finland. *European Planning Studies*, 29(10), 1906–1924. <https://doi.org/10.1080/09654313.2021.1894097>
- Przebylłowicz, E., Cunha, M. A., Geertman, S., Leleux, C., Michels, A., Tomor, Z., William, C., Webster, R., & Meijer, A. (2022). Citizen participation in the smart city: Findings from an international comparative study. *Local Government Studies*, 48(1), 23–47.
- Raudla, R. (2021). Politicians’ use of performance information in the budget process. *Public Money & Management*, 42(3), 144–151. <https://doi.org/10.1080/09540962.2021.1989779>
- Roberts, N. (2000). Wicked problems and network approaches to resolution. *International Public Management Review*, 1(1), 1–19.
- Ruhlandt, R. W. S. (2018). The governance of smart cities: A systematic literature review. *Cities*, 81, 1–23. <https://doi.org/10.1016/j.cities.2018.02.014>
- Saha, D. (2009). Factors influencing local government sustainability efforts. *State and Local Government Review*, 41(1), 39–48.
- Saha, D., & Paterson, R. G. (2008). Local government efforts to promote the “Three Es” of sustainable development: Survey in medium to large cities in the United States. *Journal of Planning Education and Research*, 28(1), 21–37. <https://doi.org/10.1177/0739456X08321803>
- Sajhau, P. (2017). IBM—building sustainable cities through partnerships and integrated approaches. *Field Actions Science Reports. The Journal of Field Actions*, (Special Issue 16), 52–57.
- Sandfort, J. (1999). The structural impediments to human service collaboration: Examining welfare reform at the front lines. *Social Service Review*, 73(3), 314–339.
- Schick, A. (1988). Micro-budgetary adaptations to fiscal stress in industrialized democracies. *Public Administration Review*, 48(1), 523–533. <https://doi.org/10.2307/975515>
- Schipper, R. P. J., & Silvius, A. J. (2018). Characteristics of smart sustainable city development: Implications for project management. *Smart Cities*, 1(1), 75–97.
- Schwartz, E. (2019). Autonomous local climate change policy: An analysis of the effect of intergovernmental relations among subnational governments. *Review of Policy Research*, 36(1), 50–74. <https://doi.org/10.1111/ropr.12320>
- Scott, T. A., & Thomas, C. W. (2017). Unpacking the collaborative toolbox: Why and when do public managers choose collaborative governance strategies? *Policy Studies Journal*, 45(1), 191–214. <https://doi.org/10.1111/psj.12162>
- Selin, H., & VanDeveer, S. D. (2011). US climate change politics and policymaking. *Wiley Interdisciplinary Reviews: Climate Change*, 2(1), 121–127.
- Sharifi, A. (2020). A typology of smart city assessment tools and indicator sets. *Sustainable Cities and Society*, 53, 101936.
- Smart Cities Council. (2015). *Smart cities readiness guide*. Smart Cities Council, 6.

- Snider, K. F., & Rendon, R. G. (2012). Public procurement: Public administration and public service perspectives. *Journal of Public Affairs Education*, 18(2), 327–348. <https://doi.org/10.1080/15236803.2012.12001687>
- Snis, U. L., Olsson, A. K., & Bernhard, I. (2021). Becoming a smart old town – How to manage stakeholder collaboration and cultural heritage. *Journal of Cultural Heritage Management and Sustainable Development*, 11(4), 627–641. <https://doi.org/10.1108/JCHMSD-10-2020-0148>
- Sørensen, E., & Torfing, J. (2021). Radical and disruptive answers to downstream problems in collaborative governance? *Public Management Review*, 23(11), 1590–1611. <https://doi.org/10.1080/14719037.2021.1879914>
- Soujja, I., Nukpezah, J. A., & Benavides, A. D. (2021). Coordination effectiveness during public health emergencies: An institutional collective action framework. *Administration & Society*, 53(7), 1014–1045. <https://doi.org/10.1177/0095399720985440>
- Stritch, J. M., Bretschneider, S., Darnall, N., Hsueh, L., & Chen, Y. (2020). Sustainability policy objectives, centralized decision making, and efficiency in public procurement processes in U.S. local governments. *Sustainability*, 12(17), 6934. <https://doi.org/10.3390/su12176934>
- Svara, J. H., Watt, T. C., & Jang, H. S. (2013). How are US cities doing sustainability? Who is getting on the sustainability train, and why? *Cityscape*, 15(1), 9–44.
- Svara, J. H., Watt, T. C., & Takai, K. (2015). Advancing social equity as an integral dimension of sustainability in local communities. *Cityscape*, 17(2), 139–166.
- Swann, W. L., & Deslatte, A. (2019). What do we know about urban sustainability? A research synthesis and nonparametric assessment. *Urban Studies*, 56(9), 1729–1747. <https://doi.org/10.1177/0042098018779713>
- Terman, J., & Smith, C. (2018). Putting your money where your mouth is: Green procurement as a form of sustainability. *Journal of Public Procurement*, 18(3), 202–216. <https://doi.org/10.1108/JOPP-09-2018-013>
- Thai, K. V., & Grimm, R. (2000). Government procurement: Past and current developments. *Journal of Public Budgeting, Accounting & Financial Management*, 12(2), 231–247.
- Thomson, A. M., & Perry, J. L. (2006). Collaboration processes: Inside the black box. *Public Administration Review*, 66(s1), 20–32. <https://doi.org/10.1111/j.1540-6210.2006.00663>
- Ting, M. M. (2011). Organizational capacity. *The Journal of Law, Economics, & Organization*, 27(2), 245–271.
- Trammell, E., Abutabenjeh, S., & Dimand, A. M. (2020). A review of public administration research: Where does public procurement fit in? *International Journal of Public Administration*, 43(8), 655–667. <https://doi.org/10.1080/01900692.2019.1644654>
- Trindade, E. P., Hinnig, M. P. F., Moreira da, Costa, E., Marques, J. S., Bastos, R. C., & Yigitcanlar, T. (2017). Sustainable development of smart cities: A systematic review of the literature. *Journal of Open Innovation: Technology, Market, and Complexity*, 3(3), 11. <https://doi.org/10.1186/s40852-017-0063-2>
- Turner, R. (2016). *Gower handbook of project management*. Routledge.
- Vangen, S., Hayes, J. P., & Cornforth, C. (2015). Governing cross-sector, inter-organizational collaborations. *Public Management Review*, 17(9), 1237–1260. <https://doi.org/10.1080/14719037.2014.903658>
- Van Slyke, D. M. (2003). The mythology of privatization in contracting for social services. *Public Administration Review*, 63(3), 296–315. <https://doi.org/10.1111/1540-6210.00291>
- Van Slyke, D. M. (2007). Agents or stewards: Using theory to understand the government-nonprofit social service contracting relationship. *Journal of Public Administration Research and Theory*, 17(2), 157–187. <https://doi.org/10.1093/jopart/mul012>
- Viale Pereira, G., Cunha, M. A., Lampoltshammer, T. J., Parycek, P., & Testa, M. G. (2017). Increasing collaboration and participation in smart city governance: A cross-case analysis of smart city initiatives. *Information Technology for Development*, 23(3), 526–553. <https://doi.org/10.1080/02681102.2017.1353946>
- Vinzant, D. H., & Vinzant, J. C. (1996). Strategy and organizational capacity: Finding a fit. *Public Productivity & Management Review*, 20(2), 139. <https://doi.org/10.2307/3380482>
- Voelz, G. J. (2010). *Contractors in the government workplace: Managing the blended workforce*. Government Institutes.
- Walker, W. E., & Chaiken, J. M. (1982). The effects of fiscal contraction on innovation in the public sector. *Policy Sciences*, 15(2), 141–165. <https://doi.org/10.1007/BF00143076>
- Wang, Q., & Peng, J. (2018). Political embeddedness of public pension governance: An event history analysis of discount rate changes. *Public Administration Review*, 78(5), 785–794. <https://doi.org/10.1111/puar.12968>
- Wang, X., Hawkins, C. V., Lebrede, N., & Berman, E. M. (2012). Capacity to sustain sustainability: A study of U.S. cities. *Public Administration Review*, 72(6), 841–853. <https://doi.org/10.1111/j.1540-6210.2012.02566>
- Warbroek, B., Hoppe, T., Bressers, H., & Coenen, F. (2019). Testing the social, organizational, and governance factors for success in local low carbon energy initiatives. *Energy Research & Social Science*, 58, 101269. <https://doi.org/10.1016/j.erss.2019.101269>
- Warner, M. E., Aldag, A. M., & Kim, Y. (2021). Privatization and intermunicipal cooperation in US local government services: Balancing fiscal stress, need and political interests. *Public Management Review*, 23(9), 1359–1376. <https://doi.org/10.1080/14719037.2020.1751255>
- Weber, E. P., & Khademian, A. M. (2008). Wicked problems, knowledge challenges, and collaborative capacity builders in network settings. *Public Administration Review*, 68(2), 334–349. <https://doi.org/10.1111/j.1540-6210.2007.00866.x>
- Whitford, A. B., & Wong, K. (2009). Political and social foundations for environmental sustainability. *Political Research Quarterly*, 62(1), 190–204. <https://doi.org/10.1177/1065912908318322>

- Williams, P. (2002). The competent boundary spanner. *Public Administration*, 80(1), 103–124. <https://doi.org/10.1111/1467-9299.00296>
- Wiig, A. (2015). IBM's smart city as techno-utopian policy mobility. *City*, 19(2–3), 258–273. <https://doi.org/10.1080/13604813.2015.1016275>
- World Commission on Environment and Development (WCED). (1987). *Our common future*. Oxford University Press, Oxford, New York, p. 383.
- Yi, H., Suo, L., Shen, R., Zhang, J., Ramaswami, A., & Feiock, R. C. (2018). Regional governance and institutional collective action for environmental sustainability. *Public Administration Review*, 78(4), 556–566. <https://doi.org/10.1111/puar.12799>
- Yigitcanlar, T. (2015). Smart cities: An effective urban development and management model? *Australian Planner*, 52(1), 27–34.
- Yigitcanlar, T., Kamruzzaman, Md., Buys, L., Ioppolo, G., Sabatini-Marques, J., da Costa, E. M., & Yun, J. J. (2018). Understanding 'smart cities': Intertwining development drivers with desired outcomes in a multidimensional framework. *Cities*, 81, 145–160. <https://doi.org/10.1016/j.cities.2018.04.003>
- Zeemering, E. S. (2018). Sustainability management, strategy and reform in local government. *Public Management Review*, 20(1), 136–153. <https://doi.org/10.1080/14719037.2017.1293148>